The hyperelliptic genus-3 GHS cover ($m = 3, r = 5$): explicit $\mathbb{F}_q$ model, Frobenius, and the transfer executed with Cantor arithmetic

2026-10-13 · Conductor T-2463 · `research/ghs_cubic_hyperelliptic_genus3_20261013/` · recorded runs `20261014T0029Z` and addendum `20261014T0227Z` (2026-10-14; base of the first commit `12331a9a7`, pre-rebase, = `f69cec000` on the PR branch; source sha256 `be4c3374…`). **Structural stage diagnostic on toy fields.** No index calculus, no relation collection, no discrete logarithm, no cost or speedup claim. Protocol frozen before the recorded run (`PROTOCOL.md`, one dated amendment written after development runs).

Question

The cover census (PR #1759) found a hyperelliptic genus-3 type ($m = 3$, $r = 5$) of GHS cover for $E/\mathbb{F}_{q^3}$. The $m = 2$ genus-3 covers built in PR #1761 are plane quartics without class arithmetic in the repository. Can the $m = 3, r = 5$ curve be written over $\mathbb{F}_q$, does its Frobenius match, and does the transfer $E_0(\mathbb{F}_{q^3}) \to \mathrm{Jac}(C)(\mathbb{F}_q)$ run as a group map?

Construction (derived, then verified)

$K = \mathbb{F}_{q^3}$, $p \in K \setminus \mathbb{F}_q$, $r_i = p^{q^i}$, $Q \in \mathbb{F}_q[x]$ monic squarefree (split $(x-a)(x-b)$ or irreducible), $F_0 = Q(x-r_0)(x-r_1)$, $F_i = F_0^{\sigma^i}$, $N = \prod(x - r_i)$. $C = K(x, y_0, y_1, y_2)$, $y_i^2 = F_i$, descent $y_0 \mapsto y_1 \mapsto y_2 \mapsto y_0$. The quotient $D = C/\iota$ is the conic $(r_2 - r_0)X_0^2 + (r_0 - r_1)X_1^2 + (r_1 - r_2)X_2^2 = 0$ in $X_0 = u_{12}u_{20}$, … with $u_{01} = y_0y_1/(Q(x - r_1))$; in trace coordinates it is defined over $\mathbb{F}_q$. A line parametrization z through an $\mathbb{F}_q$-point gives $x = x_n(z)/x_d(z)$ and $w = y_0y_1y_2x_d(z)/(QN)$ with $w^2 = h(z) = x_n^2 + c_1x_nx_d + c_0x_d^2$ (since $F_0F_1F_2 = Q^3N^2$). Split arm: the unique $\mathbb{F}_q$-point of D over a at $z = \infty$ gives $\deg h = 7$. Transfer $\varphi(P) = \sum_{j=0}^{2} \mathrm{Fr}^j(\pi_0^*[P] - 4\infty)$, base $O = (r_0, 0)$. Derived: $\pi_{0*} \circ \varphi = [4]$, so $\ker\varphi \subset E_0(K)[4]$; the unramified subcover $K(E_0)(u_{12})$ puts $T_1 = [(r_1, 0)] - [O]$ in $\ker\varphi$.

Result (20 instances, $q = 5, 7, 11, 13$; 0 failures)

P1 descent	$\mathbb{F}_q$ coefficients 20/20; on 200 random points per instance the map lands on $w^2 = h(z)$, recovers x, commutes with Frobenius and inverts (4000/4000 each)
P2 genus 3	$\deg x(z) = 4$, h squarefree of degree 7 (split) / 8 (conjugate), 20/20
P3 Frobenius	$N_1, N_2, N_3 = q+1, q^2+1, q^3+1-3t_E$ on the $\mathbb{F}_q$ model, equal to the fibre count on the K-model; $L(T) = 1 - t_ET^3 + q^3T^6$, $L(1) = \#E_0(K)$, 20/20; Mumford-pair enumeration gives the same order, 12/12 split
P4 transfer (split)	$\varphi(P)$ reduced over $\mathbb{F}_q$ for all points; homomorphism 400/400 pairs and 100/100 doublings per instance; $\varphi(G) \neq 0$, $\ell\varphi(G) = 0$ (12/12); $\lvert\ker\varphi\rvert \in \{2, 4\}$, $\subset E_0(K)[4]$, $T_1 \in \ker$ (12/12); $\pi_{0*}\varphi(P) = 4P$ for every $P \in E_0(K)$ (12/12)
P5 census	$q = 5, 7$: census class of E_0 and both conjugates has min genus 3, type $m = 3, r = 5$ hyperelliptic, same order (11 instances); $q = 11, 13$ not in the census sample. All 167,472 enumerated candidates have $4 \mid \#E_0(K)$

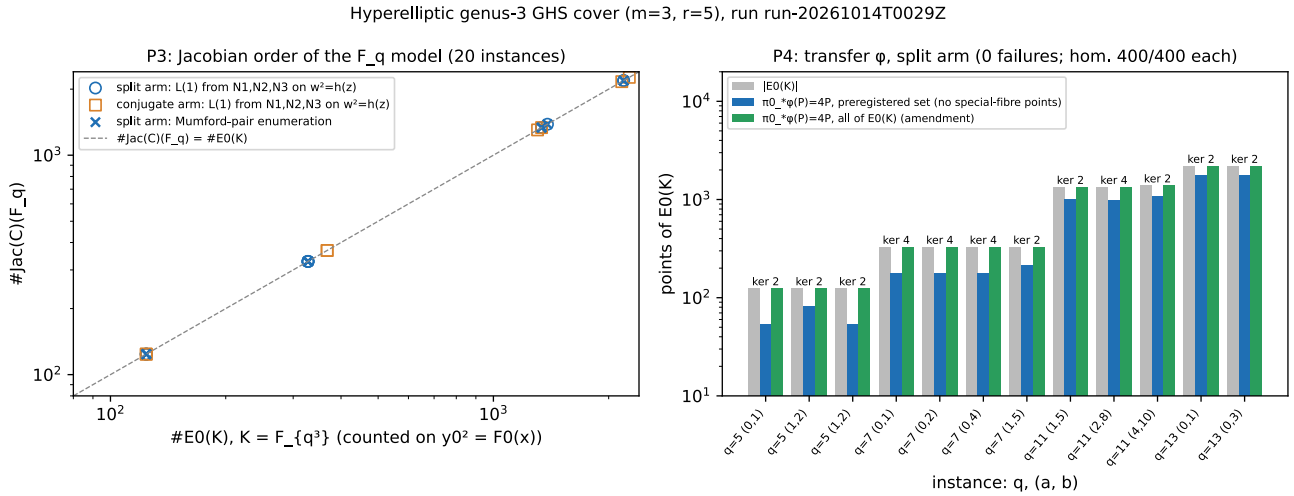


Figure 1: Run `20261014T0029Z`. Left: $\#\mathrm{Jac}(C)(\mathbb{F}_q)$ from $L(1)$ (20 instances) and from Mumford enumeration (12 split) against $\#E_0(K)$. Right: transfer checks per split instance; kernel order annotated.

Example over $\mathbb{F}_7$

$K = \mathbb{F}_7[t]/(t^3 + 3)$, $p = t$, $Q = x(x - 1)$, $\#E_0(K) = 328 = 8 \cdot 41$, $t_E = 16$: $C : w^2 = 6z^7 + z^6 + 3z^5 + 2z^4 + 2z^3 + 5z^2 + 6z + 6$, $x(z) = (z^3 + 5z^2 + 6z + 6)/(z^4 + 2z^3 + 5z^2 + 6z + 5)$. $N_k = 8, 50, 296$; $\#\mathrm{Jac}(C)(\mathbb{F}_7) = 328$. $\varphi(G) = (z^3 + 2z^2 + 5z + 4, 4z^2 + 2z + 3)$ with $41\varphi(G) = 0$; $\ker\varphi$ cyclic of order 4, image of size 82.

Addendum 2026-10-14 (run `20261014T0227Z`): conjugate-arm transfer, census at $q \geq 11$

Addendum protocol frozen before the runs, with one dated correction written after a development run (the reduction-step bookkeeping at infinity). The run reruns all 20 instances; the split-arm records equal those of `20261014T0029Z`. The failed attempt `20261014T0223Z` (a point over the original $z = \infty$ was not mapped) is kept.

Model. The conjugate-arm Weierstrass points lie over $a, b \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$, so there is no degree-7 $\mathbb{F}_q$-model. Moving an $\mathbb{F}_q$-point (z_0, w_0) to infinity gives a monic degree-8 f with $\infty_\pm = (z_0, \pm w_0)$. Arithmetic uses balanced classes $\mathrm{div}(u, v) + n\infty_+ + (3 - \deg u - n)\infty_- - D_\infty$ with $D_\infty = 2\infty_+ + \infty_-$ (Galbraith–Harrison–Mireles), and $\varphi(P) = [\sum Q_k - 6(\infty_+ + \infty_-)]$.

V validation (8 inst.)	enumerating reduced triples gives $\#E_0(K)$ (8/8); associativity 300/300, $[\#E_0(K)]D = 0$ 100/100, inverse and identity 100/100, per instance
P4c transfer (8 inst.)	valid reduced $\mathbb{F}_q$ triples for all points; homomorphism 400/400 and doublings 100/100 per instance; $\varphi(G) \neq 0$, $\ell\varphi(G) = 0$ ($\ell = 31, 31, 23, 23, 167, 163, 283, 541$; cofactors $4, 4, 16, 16, 8, 8, 8, 4$); $\lvert\ker\varphi\rvert = 2$ (3) or 4 (5, cyclic), $\subset E_0(K)[4]$, $T_1 \in \ker$; $\pi_{0*}\varphi(P) = 4P$ on every point, special fibres included, 0 failures, 0 not evaluable
P5c census $q = 11, 13$	census fork reproduces the merged census at $q = 5, 7$ byte for byte (976 lines); over all 1,786,323 / 4,855,539 structures, all 27 curves (9 E_0 × 3 conjugates) have minimal genus 3, the type $m = 3, r = 5$ hyperelliptic, the same order, 0 mismatches. Genus-3 structures per curve: split 16–24 (4 hyperelliptic, the rest $m = 2$); conjugate exactly 2, both hyperelliptic

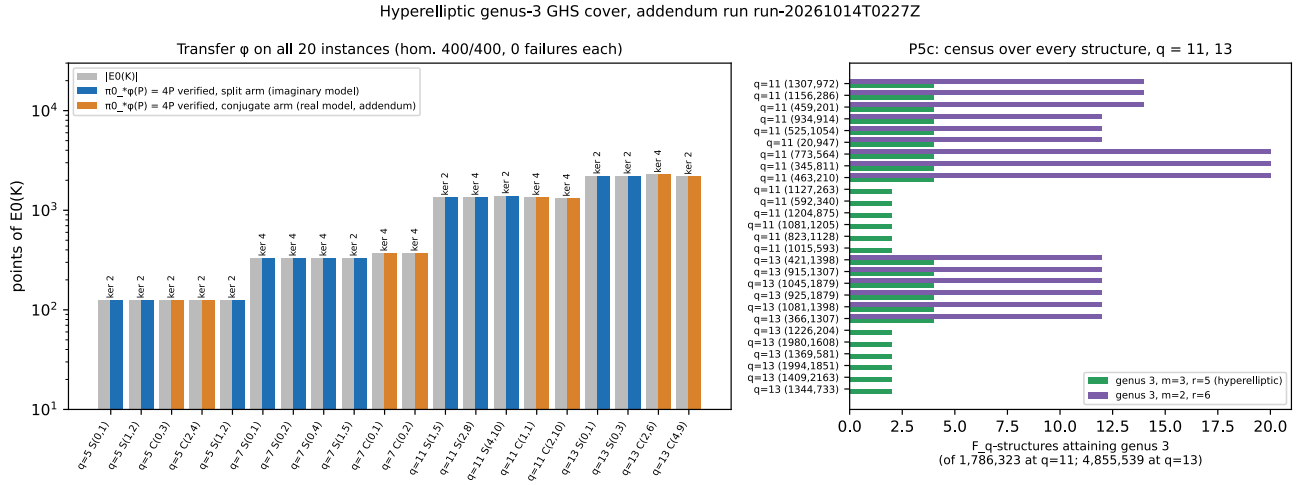


Figure 2: Run `20261014T0227Z`. Left: $\pi_{0*}\varphi(P) = 4P$ verified on every point of $E_0(K)$ for all 20 instances (S split, C conjugate), kernel order annotated. Right: genus-3 structures per curve in the direct census at $q = 11, 13$.

Lean

`lean/HyperGenus3.lean` (v4.34.1, no Mathlib, no `sorry`): branch-set combinatorics ($r = 5$, $m = 3$, genus 3, rank-2 subspace with union of size 3, $\tau B \Delta \tau^2 B \subset B$, triple combination $= \{a, b\}$), the conic identities, $F_0F_1F_2 = Q^3N^2$, $w^2 = h$ multiplied out; addendum: the norm identity and the integer bookkeeping at infinity of the real-model steps, composition, negation and the $6(\infty_+ - \infty_-)$ correction. Not formalized: function-field descent, Frobenius, L-polynomial, the transfer, $\pi_{0*}\varphi = [4]$, uniqueness of balanced representatives (numerical only).

Limits

Closes the plane-quartic obstruction **by a different route for this type only**; plane-quartic class arithmetic is still missing. φ is not an isomorphism (index 2 or 4) on either arm. The real-model arithmetic is study-local and genus-3 only. Toy fields, first instances in a fixed order. No cost statement.